

3.10.5 X-ray imaging (A-level only)

3.10.5.1 The physics of diagnostic X-rays(A-level only)

Content

Physical principles of the production of X-rays; maximum photon energy, energy spectrum; continuous spectrum and characteristic spectrum.

Rotating-anode X-ray tube; methods of controlling the beam intensity, the photon energy, the image sharpness and contrast, and the patient dose.

3.10.5.2 Image detection and enhancement (A-level only)

Content

Flat panel (FTP) detector including X-ray scintillator, photodiode pixels, electronic scanning.

Advantages of FTP detector compared with photographic detection.

Contrast enhancement; use of X-ray opaque material as illustrated by the barium meal technique.

Photographic detection with intensifying screen and fluoroscopic image intensification; reasons for using these.

Exponential attenuation.

Linear coefficient μ , mass attenuation coefficient μ_m , half-value thickness

$$I = I_0 e^{-\mu x} \quad \mu_m = \frac{\mu}{\rho}$$

Differential tissue absorption of X-rays excluding details of the absorption processes.

3.10.5.4 CT scanner (A-level only)

Content

Basic principles of CT scanner:

- movement of X-ray tube
- narrow, monochromatic X-ray beam
- array of detectors
- computer used to process the signals and produce a visual image.

Comparisons will be limited to advantages and disadvantages of image resolution, cost and safety issues. Students will not be asked about the construction or operation of the detectors.